

Linear TIME experiment guideline

1 Question

Linear TIME measures how exact ridge forecasting changes with temporal scale, dataset semantics, user generalization, channel interaction, normalization, loss, detrending and constant windows. The implementation uses no foundation model or learned iterative optimizer.

2 Data and tasks

TIME's saved-Arrow targets retain item and channel identities. The project YAML declares whether these identities represent distinct physical quantities, repeated users/sensors, or both. One task is one physical-quantity panel, one L-H cell, one model mode and one scientific configuration. TIME chronological validation/test lengths remain fixed. Proposed L is capped by history available at the earliest test origin. Validation walks backward from the first test origin at stride H, uses at most the test-date count, and keeps only finite L-input/H-output windows. Zero usable validation windows selects the configured default alpha, while training still requires at least one valid window. Proposed H is capped in the YAML when a TIME evaluation interval is shorter.

Front	subset	models	seeds
01 default	all retained semantic panels	joint	none
02 user generalization	repeated-member panels	shared	0,1,2
03 variate modes	panels with $N \geq 2$	independent, joint, shared	none

Every front runs all nine dataset-specific L-H cells. The default and variate experiments use every member. User generalization partitions members 80/20 and reports seen/unseen validation/test populations.

3 Execution

Each root Slurm front resolves the checkout through `SLURM_SUBMIT_DIR`, loads project-owned runtime paths, records one compute-node resource snapshot, and calls `src/slurm/study.sh`. That shell runs fit/evaluate and report stages sequentially. Task manifests are allocated before fitting and own exact configuration/recovery identity. Paths contain readable factors and optional seed folders; they contain no hashes.

For each task, pseudocode is:

```
load Arrow targets and semantic YAML
split distinct quantities; preserve repeated entities in a panel
effective L = min(proposed L, history at earliest test origin)
stream training panel windows and prepare exact ridge system
for alpha in validation grid: solve reused system; score validation
solve selected alpha on training statistics
evaluate ridge, persistence, Seasonal Naive
write state, window metrics, summaries, timings, manifest
```

Training statistics, validation alpha choice, selected coefficients, and each population/split evaluation are separate exact caches. Therefore a changed alpha grid reuses training statistics, and unchanged selected coefficients reuse their evaluation products.

Reports consume exactly the completed task manifests named by their launch. All metrics keep finite support and matched Seasonal denominators explicit. Only seeded user generalization reports sample SD across partitions. Every PNG is paired with a PDF and all artifacts are enumerated by the report manifest.

4 Current evidence

No real TIME execution has been submitted or analyzed. The maintained synthetic checkpoint covers semantic configuration, all three exact model modes, the L cap, seasonal evaluation, seed scope, task lifecycle, plots and Slurm headers.